# CBE 30355 Transport I Midterm Exam

**October 15, 2026**

Closed Books and Notes: Non-graphing calculator only

## Problem 1 (20 points)

In class we demonstrated the fascinating phenomenon of the "tears of wine". The fluid making up the tears is provided by a very thin film flowing upwards via a surface tension gradient driven by the evaporation of alcohol. In this problem we prove that this film must be too thin to be visible via an unaided eyeball (e.g., less than about $100\mu m$). Consider a thin viscous film on a **vertical wall**. The film is driven upward by a Marangoni stress $\frac{d\Gamma}{dx}$ at its free surface and retarded by gravity, with no-slip at the wall. If the film is sufficiently thin, the Marangoni stress wins and the flow is upwards, but if it is too thick gravity wins and the net flow is downwards. Thus, the maximum possible film thickness (since there must be at least some upward flow to fuel the tears) is where the net flow is zero.

**(a)** Assuming unidirectional flow, write down the equations and boundary conditions governing the flow in the thin film.

**(b)** Taking the film thickness to be some unknown value $\delta$, render the problem dimensionless.

**(c)** Solve the equations to determine the flow rate as a function of $\delta$ and the parameters of the problem, and develop the equation for the value of $\delta$ for which the net flow is zero.

**(d)** For the cooking sherry used in the demonstration ($\rho \approx 1$ g/cm$^3$, $\mu = 0.015$ g/(cm s), $g = 980$ cm/s$^2$) and a surface tension gradient of $d\Gamma/dx \approx 2$ dyne/cm$^2$, calculate this maximum value of $\delta$.

## Problem 2 (15 points)

During extreme flood events, such as the record February 2018 flood on the St. Joseph River, peak flow rates passing over the Twin Branch Dam in Mishawaka reach $Q = 600\,\mathrm{m}^3/\mathrm{s}$ across its multi-bay spillway structure, which has a total effective width of $W = 100\,\mathrm{m}$.

A primary hydraulic safety concern is whether the spillway training walls and dividing piers are tall enough to contain the flow during high discharge, especially during asymmetric gate operation scenarios where cross-waves and air entrainment increase the local water depth $D$. Because these complex flow effects cannot easily be calculated from first principles, engineers validate chute wall heights experimentally using scale models.

It is proposed to evaluate these operational flood scenarios by constructing a geometrically similar scale model at $\lambda = 1 : 40$ (all lengths in the model are 1/40 of the full-scale dam) and measuring the water depth $D_m$ in a laboratory flume.

**(a)** What is the appropriate dimensionless parameter to preserve in the model test, and why? What parameter is *not* preserved, and why is this acceptable (for now)?

**(b)** What flow rate $Q_m$ must be supplied to the model to achieve dynamic similarity? If the model flow depth is measured to be $D_m$, what is the corresponding full-scale spillway depth $D$?

**(c)** Show that the Reynolds number can be estimated without knowing the flow depth $D$. Using kinematic viscosity $\nu = 10^{-6}\,\mathrm{m}^2/\mathrm{s}$, determine $Re$ for both the prototype and the model. Determine the minimum full-scale flow rate $Q$ for which this $\lambda = 1:40$ model could still maintain approximate dynamic similarity (i.e., $Re_m \geq 10^4$).



Figure 1: Spillway section of the Twin Branch Dam on the St. Joseph River in Mishawaka, IN. High-velocity flood flows must be safely contained within the training walls during peak discharge events.

## Problem 3 (15 points)

You are trying to put out a fire in a building at a height $H$ by squirting water up from street level. You have a pump which is capable of supplying water with a useful mechanical energy/time (e.g., effective pumping power) of $P_W$. For this problem, neglect all losses (e.g., use Bernoulli's Equation!).

**a.** You want to put the fire out quickly, so you want a large flow rate $Q$. For the parameters of the problem, what diameter nozzle $D$ will achieve the maximum flow rate?

**b.** In part (a) the jet was going straight up - but you find that you can't get the fire truck that close to the building! If the jet instead has to be launched from a 45 degree angle, by what fraction is the maximum flow rate on the target and necessary nozzle diameter reduced?

## Problem 4 (10 points)

In a class demonstration of Archimedes law the scale read 0.65 g when my wedding ring was dangled in a tared beaker of water and 7.52 g when it was allowed to settle to the bottom. If the density of gold is $19.3\,\mathrm{g/cm}^3$ and the density of the alloying metal (mostly copper, and a little silver) is $9.0\,\mathrm{g/cm}^3$, what is the karat of the ring? Recall that this is on a mass basis, and pure gold is 24K.